Linear Algebra
MATH 2250-03

Name (Print): ANSWER KEY

- This exam contains 8 pages (including the front and back of this cover page) and 8 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have 105 minutes (5:15 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 220 points total in this exam, as well as 1 bonus questions worth a total of 20 possible points.

Page:	3	4	5	6	Total
Points:	20	60	60	80	220
Score:					

Bonus:	

1. Mark each statement as $\boxed{\text{True}}$ or $\boxed{\text{False}}$ by completely shading in the box corresponding to your answer.

   For readability in the key, correct answers are colored in red, e.g. Correct Answer, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

   (a) $\boxed{\text{2 points}}$ Elementary row operation on an augmented matrix never change the solution set of the associated linear system.

   True ~~False~~

   (b) $\boxed{\text{2 points}}$ The reduced row echelon form of a matrix is unique.

   True ~~False~~

   (c) $\boxed{\text{2 points}}$ A homogeneous system is always consistent.

   True ~~False~~

   (d) $\boxed{\text{2 points}}$ If A, B, and C are matrices such that $AB = AC$, then $B = C$.

   ~~True~~ False

   (e) $\boxed{\text{2 points}}$ The columns of any 4×5 matrix are linearly dependent.

   True ~~False~~

   (f) $\boxed{\text{2 points}}$ Every matrix transformation is a linear transformation.

   True ~~False~~

   (g) $\boxed{\text{2 points}}$ A linear transformation preserves the operations of vector addition and scalar multiplication.

   True ~~False~~

   (h) $\boxed{\text{2 points}}$ If two vectors are linearly independent, then they both lie on the same line through the origin.

   ~~True~~ False

   (i) $\boxed{\text{2 points}}$ If A and B are $n \times n$ matrices such that $AB = I$, then A and B are invertible.

   True ~~False~~

   (j) $\boxed{\text{2 points}}$ A column of the standard matrix for a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is the image of a column of the $n \times n$ identity matrix.

   True ~~False~~

2. $\boxed{\text{30 points}}$ Determine the value(s) of $h \in \mathbb{R}$ such that the following matrix is the augmented matrix of a consistent linear system:

$$\begin{bmatrix} 2 & 3 & \cos(h) \\ 4 & 6 & \sqrt{2} \end{bmatrix}$$

Elementary row operations allow us to carry out the following reduction on the augmented matrix:

$$\begin{bmatrix} 2 & 3 & \cos(h) \\ 4 & 6 & \sqrt{2} \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 3 & \cos(h) \\ 0 & 0 & \sqrt{2} - 2\cos(h) \end{bmatrix} \quad (\text{Replace } R_2 \text{ with } R_2 - 2R_1)$$

Note that the system corresponding to the augmented matrix in the last line of the above calculations is

$$2x_1 + 3x_2 = \cos(h)$$
$$0 = \sqrt{2} - 2\cos(h)$$

Clearly, this system is inconsistent unless $\cos(h) = \frac{\sqrt{2}}{2}$. Moreover, as x_2 is a free variable, the first equation in this system has a solution for any value of h. Thus, we may conclude that the only value of h (in $[0, \pi)$, at least) for which the augmented matrix given in the statement of the problem corresponds to a consistent linear system is $h = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$.

3. $\boxed{\text{30 points}}$ Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$. Do the columns of A^T span $\mathbb{R}^2$? Justify your answer.

We seek to determine whether the columns of A^T span$\mathbb{R}^2$. That is, if $x \in \mathbb{R}^2$ is an arbitrary vector, we want to know if we can always find constants $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

This amounts to asking whether $A^T x = b$ has a solution for every $b \in \mathbb{R}^2$. So, we row-reduce A^T and produce:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \overset{\text{Row Reduce}}{\Rightarrow} \begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{bmatrix}$$

As this clearly shows that A^T has a pivot in every row, it follows that $A^T x = b$ has a solution for all $b \in \mathbb{R}^2$. Therefore, we may conclude that the columns of A^T do indeed span $\mathbb{R}^2$.

4. $\boxed{\text{30 points}}$ Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation such that

$$T(x_1, x_2) = \begin{bmatrix} x_1 - 2x_2 \\ -x_1 + 3x_2 \\ 3x_1 - 2x_2 \end{bmatrix}.$$

Find the standard matrix corresponding to the linear transformation T.

We seek the standard matrix corresponding to the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ given above. To that end, recall that the standard matrix of a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ is given by $\begin{bmatrix} L(e_1) & \dots & L(e_n) \end{bmatrix}$, where e_i denotes the i^{th} column of the $n \times n$ identity matrix. As such, the standard matrix corresponding to T is given by $\begin{bmatrix} T(e_1) \\ T(e_2) \end{bmatrix}$. Computing, we produce

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 - 2(0) \\ -1 + 3(0) \\ 3 - 2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 - 2 \\ -0 + 3 \\ 3(0) - 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}$$

So, the standard matrix for T is $\begin{bmatrix} 1 & -2 \\ -1 & 3 \\ 3 & -2 \end{bmatrix}$.

5. $\boxed{\text{30 points}}$ Let $A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$. Give an explicit formula for a general solution for the inhomogeneous system $Ax = b$ without using back-substitution or row-reduction.

We seek a general solution to the inhomogeneous system $Ax = b$. To that end, let us recall that a 2×2 matrix M of the form $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $ad - bc \neq 0$ is invertible and $M^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. So, observe that $(3)(4) - (2)(7) = 12 - 14 = -2 \neq 0$. Thus, we may conclude A is invertible, and we may compute $A^{-1} = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix}$. So, since setting $x = A^{-1}b$ yields $Ax = AA^{-1}b = Ib = b$, it follows that $x = A^{-1}b$ is a general solution to $Ax = b$. Therefore, we reach our final answer by computing

$$x = (A)^{-1}b = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix}\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

6. $\boxed{\text{40 points}}$ Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is a one-to-one (i.e. T is injective) linear transformation. What relationship between n and m must be true?

Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation which is one-to-one (i.e. T is injective). We seek a relationship between n and m. To that end, let us recall the following theorem given in lecture (Theorem 12 of section 1.8 in your book):
Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$ be the standard matrix for T. Then:

1. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (i.e. T is surjective) if and only if the columns of A span $\mathbb{R}^m$.
2. T is one-to-one (i.e. T is injective) if and only if the columns of A are linearly independent.

So, as T is one-to-one by assumption, it must be the case that the columns of the matrix $A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$ are linearly independent. That is, $\{T(e_1), \dots, T(e_n)\}$ must be a linearly independent set of vectors in $\mathbb{R}^m$. But, since any set of k vectors in $\mathbb{R}^m$ is linearly independent if $k > m$, this tells us that $n \le m$ must hold.

Therefore, we may conclude that if $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation which is one-to-one (i.e. T is *injective*), then it must be the case that $n \le m$.

7. Let A be an $n \times n$ matrix, and let the linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ be defined by $T(x) = Ax$. Suppose $T(u) = T(v)$ for some $u, v \in \mathbb{R}^n$ with $u \neq v$.

(a) $\boxed{\text{20 points}}$ Does the equation $Ax = 0$ have a nontrivial solution? Justify your answer.

We seek to determine whether the homogenous system $Ax = 0$ has a nontrivial solution. To that end, note that $T(x) = T(y)$ for some $u, v \in \mathbb{R}^n$ with $u \neq v$ by assumption. So, we have

$$T(u) - T(v) = Au - Av = A(u - v) = 0.$$

As $u \neq v$, it follows that $u - v \neq 0$. Hence, $x = u - v$ is a nontrivial solution to the homogeneous system $Ax = 0$.

(b) $\boxed{\text{20 points}}$ Does the equation $Ax = z$ have a unique solution for any $z \in \mathbb{R}^n$? Justify your answer.

We seek to determine whether $Ax = z$ has a unique solution for any $z \in \mathbb{R}^n$. To that end, observe that our solution to the previous question yields that $x = u - v$ is a nontrivial solution to the homogeneous system $Ax = 0$. So, if $Ax = z$ has a solution $x = y$ for some $z \in \mathbb{R}^n$, then $x = y + u - v$ is also a solution, as

$$A(y + u - v) = Ay + A(u - v) = Ay = z.$$

Since $u - v \neq 0$, $y + u - v \neq y$, the solution to $Ax = z$ is not unique. As we chose $z \in \mathbb{R}^n$ arbitrarily, we may conclude that $Ax = z$ may not have a unique solution for any $z \in \mathbb{R}^n$.

8. $\boxed{\text{20 points (bonus)}}$ Recall that a pair of complex numbers $z_1 = x_1 + ix_2$ and $z_2 = y_1 + iy_2$ satisfy

$$z_1 z_2 = \big(x_1 + ix_2\big)\big(y_1 + iy_2\big) = \big(x_1y_1 - x_2y_2\big) + \big(x_1y_2 + x_2y_1\big)i.$$

Let $\mathbb{C}$ denote the complex numbers and let $\mathbb{M}_2(\mathbb{R})$ denote the set of 2×2 matrices with real entries. To solve this problem, find a pair of 2×2 matrices A and B such that, for any complex numbers $z_1, z_2 \in \mathbb{C}$, the map $T : \mathbb{C} \to \mathbb{M}_2(\mathbb{R})$ defined by $T(z) = T(a + ib) = aA + bB$ satisfies

$$T\big(z_1z_2\big) = T(z_1)T(z_2)$$

Since T is linear, this shows that the complex numbers have a representation as matrices, and that this representation preserves both the additive and multiplicative structure of the complex numbers.

HINT: Try using a matrix B of the form $B = \begin{bmatrix} b_1 & -b_2 \\ b_2 & b_1 \end{bmatrix}$ for some real numbers b_1 and b_2.

NOTE: There was a minor typo in this problem (see the remark after the answer given below)

If $T(z_1z_2) = T(z_2)T(z_2)$, then we have

$$\begin{aligned} T(z_1z_2) &= T(z_1)T(z_2) \\ \big(x_1y_1 - x_2y_2\big)A + \big(x_1y_2 + x_2y_1\big)B &= \big(x_1A + x_2B\big)\big(y_1A + y_2B\big) \\ \big(x_1y_1 - x_2y_2\big)A + \big(x_1y_2 + x_2y_1\big)B &= x_1y_1A^2 + x_2y_2B^2 + x_2y_1BA + x_1y_2AB. \end{aligned}$$

Matching the coefficients up across the equality, we can see that if $BA = AB = B$, $A^2 = A$ and $B^2 = -A$, then we have

$$x_1y_1A^2 + x_2y_2B^2 + x_2y_1BA + x_1y_2AB = \big(x_1y_1 - x_2y_2\big)A + \big(x_1y_2 + x_2y_1\big)B.$$

This is what we want, so we only need to figure out what A and B should be for this to happen. Since $AB = BA = B$, and $A^2 = A$, it's clear that we need to set $A = I$. So, we then only need to figure out a matrix B such that $B^2 = -I$. Taking $B = \begin{bmatrix} b_1 & -b_2 \\ b_2 & b_1 \end{bmatrix}$ as the hint suggests, we have

$$B^2 = -I$$
$$\begin{bmatrix} (b_1^2 - b_2^2) & (-2b_1b_2) \\ (2b_1b_2) & (b_1^2 - b_2^2) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

So, $b_1^2 - b_2^2 = -1$, $-2b_1b_2 = 0$, and $2b_1b_2 = 0$. From the equations $-2b_1b_2 = 2b_1b_2 = 0$, either $b_1 = 0$, $b_2 = 0$, or both b_1 and b_2 are zero. From the equation $b_1^2 - b_2^2 = -1$, clearly we can't have both b_1 and b_2 be zero. Further, since a real number squared is non-negative, we need to choose $b_1 = 0$ and $b_2 = 1$ to satisfy $b_1^2 - b_2^2 = -1$. So, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and

$$T(z) = T(x + iy) = x\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + y\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

REMARK:
If you didn't hear my announcement during the exam about the typo, the problem as it was printed on your exams stated that

"A pair of complex numbers $z_1 = x_1 + ix_2$ and $z_2 = y_1 + iy_2$ satisfy

$$z_1 z_2 = (x_1 + ix_2)(y_1 + iy_2) = (x_1y_1 - x_2y_2) + (x_1y_2 - x_2y_1)i."$$

The error was the minus sign between the terms x_1y_2 and x_2y_1 in the coefficient of i (it should have been a plus). Unfortunately, this changes things, and would probably have prevented most of you from finding a solution (we haven't covered the materials and tools that you would need to employ to deal with problems of this form). So, to compensate, *anyone who attempted the extra credit problem at all was awarded half points.* In some instances where it was apparent that a substantial effort had been made to solve the version of the problem with the typo, further points beyond half were awarded on a case-by-case basis.